

| Dataset                          | $\#F_{total}$ | $\min( Var(F) )$ | $\max( Var(F) )$ | $\min( F )$ | $\max( F )$ |
|----------------------------------|---------------|------------------|------------------|-------------|-------------|
| Global                           | 4550          | 0                | 8286433          | 0           | 7689680     |
| $\hookrightarrow$ Lagniez [?]    | 1948          | 5                | 229100           | 10          | 399794      |
| $\hookrightarrow$ Soos [?]       | 1823          | 2                | 8286433          | 1           | 7689680     |
| $\hookrightarrow$ Plazar [?]     | 501           | 14               | 486193           | 31          | 2598178     |
| $\hookrightarrow$ Sundermann [?] | 278           | 0                | 31012            | 0           | 350120      |

Table 1: Dataset summary. The first column indicates the dataset, and the  $\#F$  column indicates how many formulae the dataset contains. The following columns indicate the minimum and maximum number of variables (resp. clauses) in the dataset.

| Dataset                          | $\#F_{total}$ | $\#D4$ | $\#\neg D4$ | $\#+D4_{ld}$ | $\#+CK$ | $\#+DivKC$ |
|----------------------------------|---------------|--------|-------------|--------------|---------|------------|
| Global                           | 4550          | 185    | 670         | 1            | 79      | 55         |
| $\hookrightarrow$ Lagniez [?]    | 1948          | 53     | 213         | 1            | 39      | 23         |
| $\hookrightarrow$ Soos [?]       | 1823          | 100    | 393         | 0            | 40      | 28         |
| $\hookrightarrow$ Plazar [?]     | 501           | 31     | 61          | 0            | 0       | 4          |
| $\hookrightarrow$ Sundermann [?] | 278           | 1      | 3           | 0            | 0       | 0          |

Table 2: Experimental results regarding the scalability of Algorithm ?? . Column  $\#F_{total}$  indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by Algorithm ?. Column  $\#\neg D4$  shows the number of formulae not compiled by D4. The last column indicates the number of formulae that were only compiled by Algorithm ??, but not by D4.

| Dataset                          | $\#F$ | $Y_l \leq  R_F $ | $Y_h \geq  R_F $ | Coverage | $ R_{G_P}  \leq  R_F  \leq  R_{G_U} $ |
|----------------------------------|-------|------------------|------------------|----------|---------------------------------------|
| Global                           | 3670  | 0.969            | 0.857            | 0.826    | 1.000                                 |
| $\hookrightarrow$ Lagniez [?]    | 1689  | 0.969            | 0.895            | 0.864    | 1.000                                 |
| $\hookrightarrow$ Soos [?]       | 1307  | 0.969            | 0.906            | 0.875    | 1.000                                 |
| $\hookrightarrow$ Plazar [?]     | 403   | 0.968            | 0.789            | 0.757    | 1.000                                 |
| $\hookrightarrow$ Sundermann [?] | 271   | 0.967            | 0.487            | 0.454    | 1.000                                 |

Table 3: Experimental results for Algorithm ?. Column  $\#F$  indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often  $|R_F|$  is within the confidence interval  $[Y_l; Y_h]$  and thus measures the accuracy of our method.

| Dataset                          | #F   | $Y_l \geq  R_{G_P} $ | $Y_h \leq  R_{G_U} $ | Both  | $median(r_c)$ | $max(r_c)$ |
|----------------------------------|------|----------------------|----------------------|-------|---------------|------------|
| Global                           | 3669 | 0.807                | 0.836                | 0.721 | 4.17e-9       | 0.54       |
| $\hookrightarrow$ Lagniez [?]    | 1689 | 0.844                | 0.869                | 0.760 | 1.8e-9        | 0.54       |
| $\hookrightarrow$ Soos [?]       | 1307 | 0.920                | 0.893                | 0.834 | 1.75e-7       | 0.0962     |
| $\hookrightarrow$ Plazar [?]     | 403  | 0.697                | 0.769                | 0.610 | 1.64e-13      | 0.0153     |
| $\hookrightarrow$ Sundermann [?] | 270  | 0.196                | 0.448                | 0.093 | 1.69e-13      | 0.281      |

Table 4: Experimental results comparing the bounds obtained with Algorithm ?? and with Lemma ??. Column #F indicates with how many formulae the statistics have been computed. The 'Both' column indicates how often we have  $Y_l \geq |R_{G_P}| \wedge Y_l \leq |R_F|$  and  $Y_h \leq |R_{G_U}| \wedge Y_h \geq |R_F|$ . The last two columns represent the observed median and maximum values of the ratio  $r_c = \frac{\min(Y_h, |R_{G_U}|) - \max(Y_l, |R_{G_P}|)}{|R_{G_U}| - |R_{G_P}|}$ , which was calculated exclusively if  $Y_l \leq |R_F| \leq Y_h$ . The number of formulae on which the last two columns are computed can easily be obtained by multiplying the #F column with the 'Coverage' column in Table 3.